

Přednáška 01

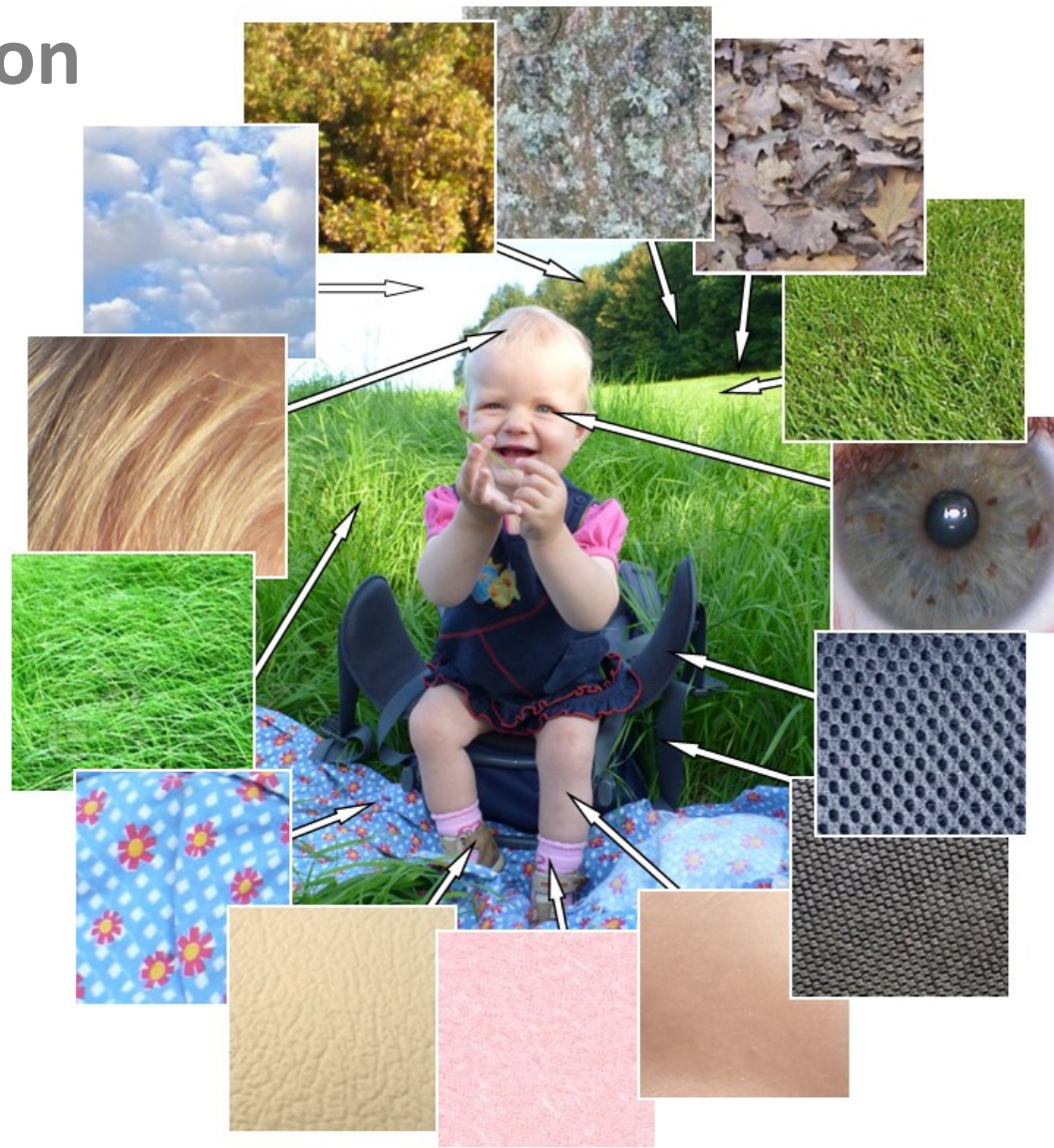
Úvod do vzhledu materiálů

Jiří Filip filipj@utia.cas.cz

UTIA, Czech Academy of Sciences

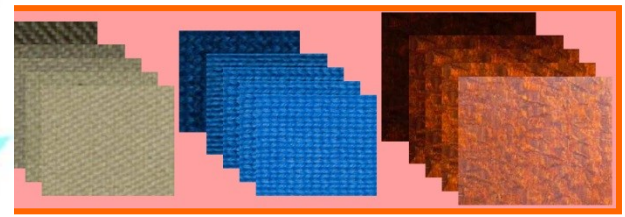
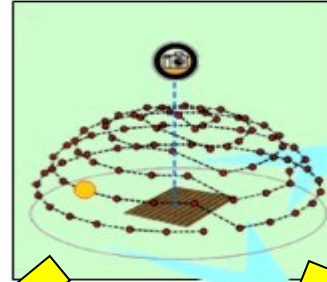
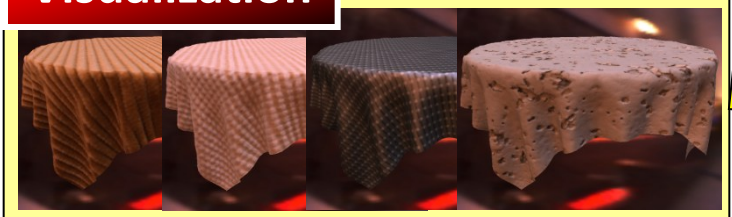


Motivation

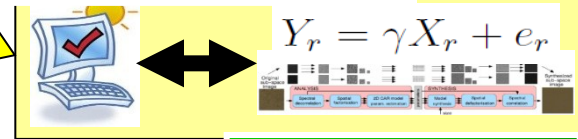


**Accurate
Appearance
Measurement**

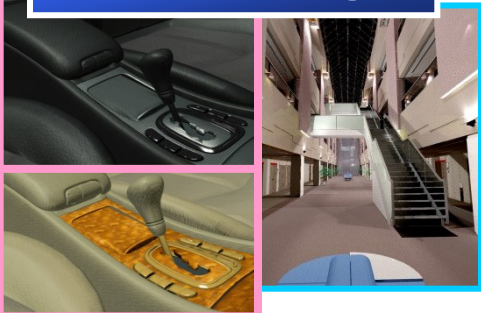
Visualization



Visual Scene Analysis



Interior Design



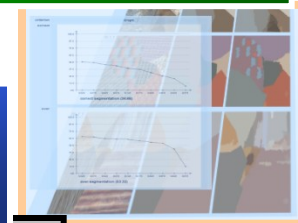
**Culture Heritage
Digitization**



**Visual Safety
Simulation**



Scene Segmentation



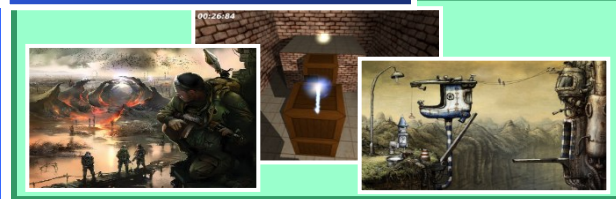
**Visual
Psychophysics**



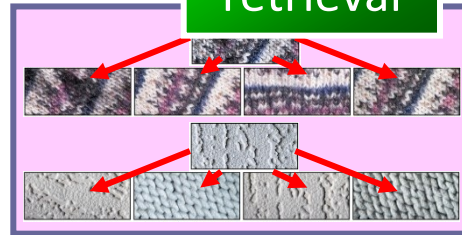
Movie Industry



Game Industry



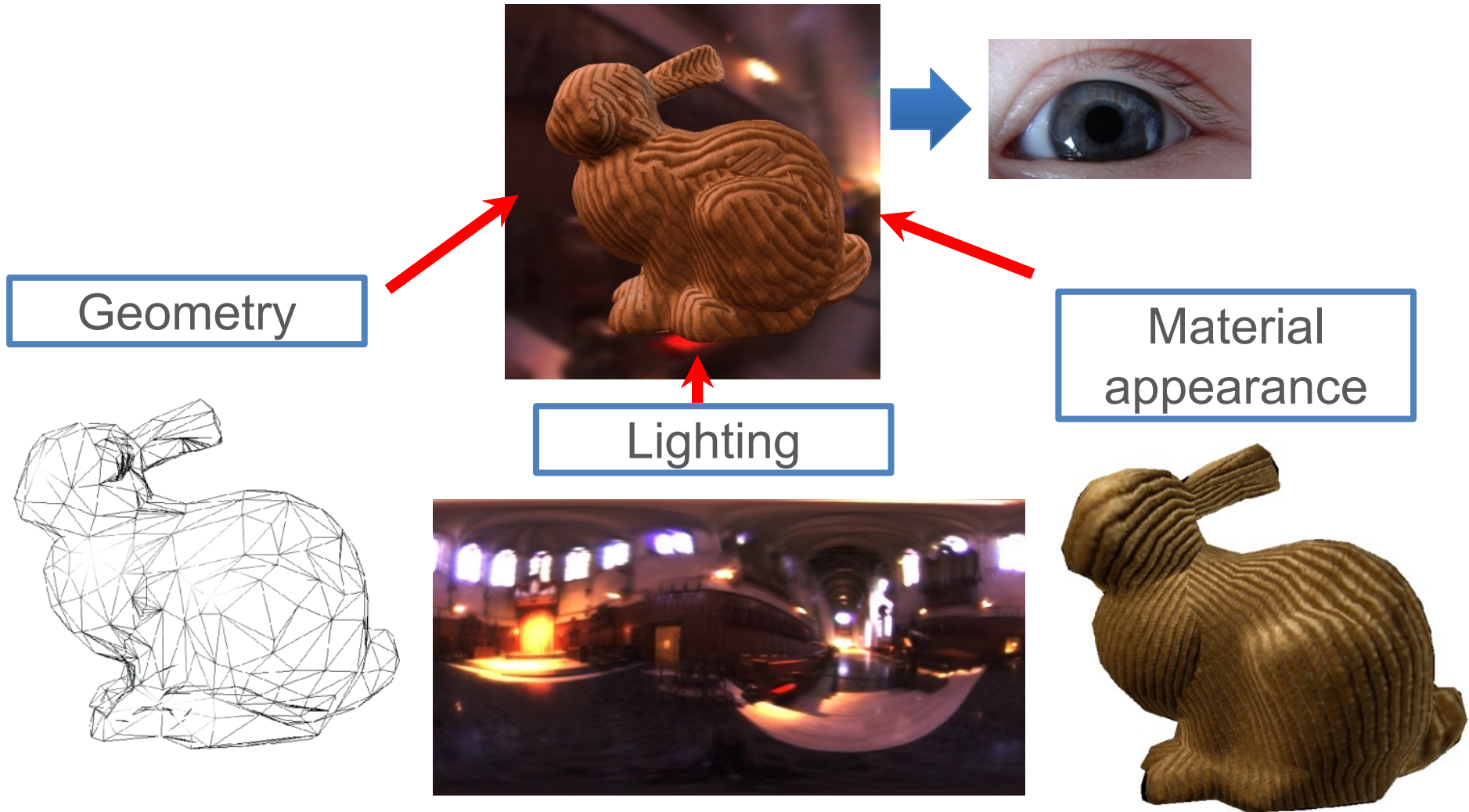
**Image
retrieval**



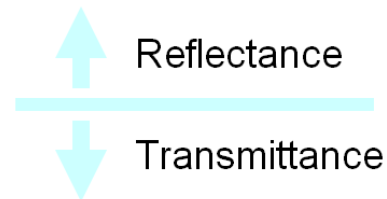
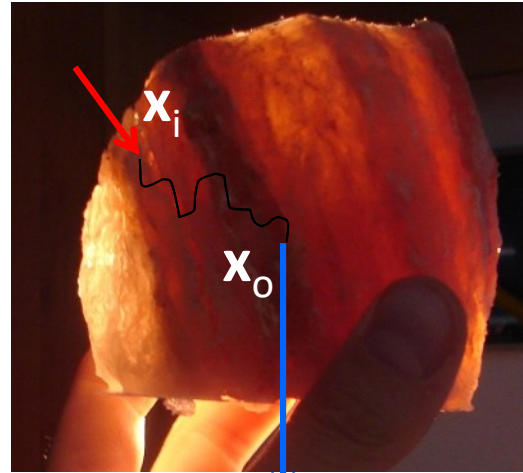
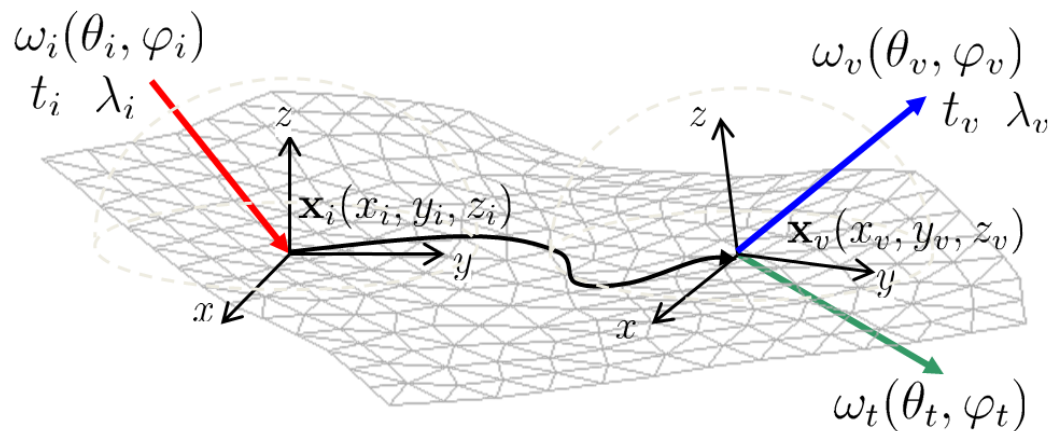
**Medical
Images
Analysis**



Digital Twin of Real-World Material



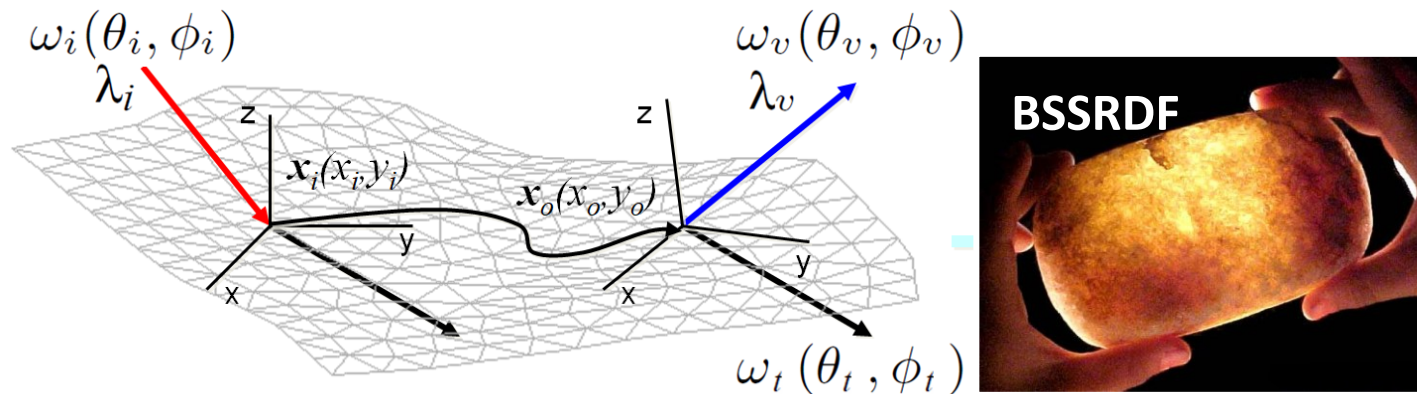
General Reflectance Function



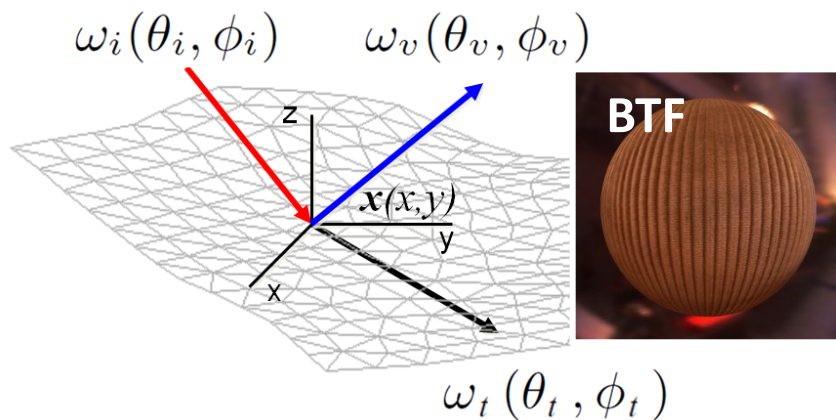
$$Y_r = GRF(\lambda_i, x_i, y_i, z_i, t_i, \theta_i, \varphi_i, \lambda_v, x_v, y_v, z_v, t_v, \theta_v, \varphi_v, \theta_t, \varphi_t)$$

16 dimensions

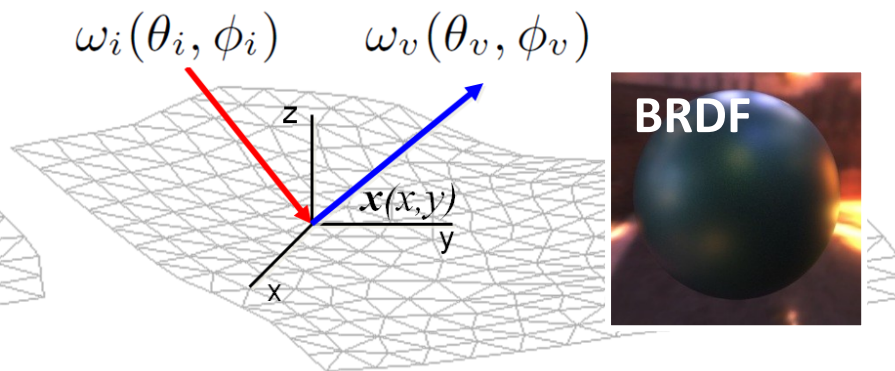
General model of light-material interaction



Without sub-surface scattering

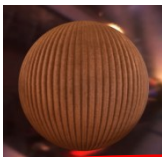


Opaque materials

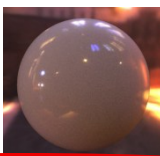


Taxonomy of Appearance Representations

Textured materials



Homogeneous materials



9D

Reflectance Fields / BSSRDF

$(\lambda, x_i, y_i, \theta_i, \varphi_i, x_v, y_v, \theta_v, \varphi_v)$

Bidirectional
surface scattering
reflectance
distribution
function

7D

Spatially varying BRDF

SVBRDF

Bidirectional texture
function

BTF

Bidirectional scattering distribution function

BSDF

$(\lambda, x, y, \theta_i, \varphi_i, \theta_v, \varphi_v)$

$(\lambda, \theta_i, \varphi_i, \theta_v, \varphi_v, \theta_t)$

5D

Surface light field

SLF

Surface reflectance field

SRF

Bidirectional reflectance / transmittance
distribution functions

BRDF

BTDF

$(\lambda, x, y, \theta_v, \varphi_v)$

$(\lambda, x, y, \theta_i, \varphi_i)$

$(\lambda, \theta_i, \varphi_i, \theta_v, \varphi_v)$

$(\lambda, \theta_i, \varphi_i, \theta_t, \varphi_t)$

4D

Dynamic Texture

Isotropic BRDF

$(\lambda, \theta_i, \theta_v, |\varphi_i - \varphi_v|)$

Texture, bump-map, albedo map, etc.

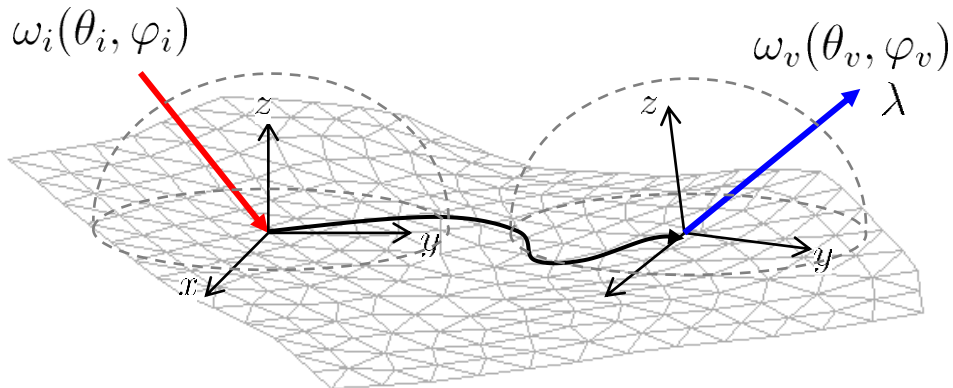
3D

Multispectral Texture

Textured Material Representation (1/2)

BSSRDF

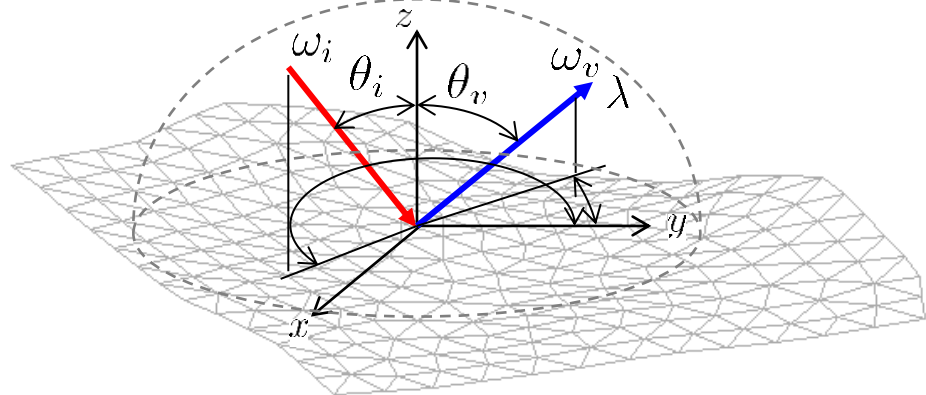
9D



$$Y_r = BSSRDF(\lambda, x_i, y_i, \theta_i, \varphi_i, x_v, y_v, \theta_v, \varphi_v)$$

BTF / SVBRDF

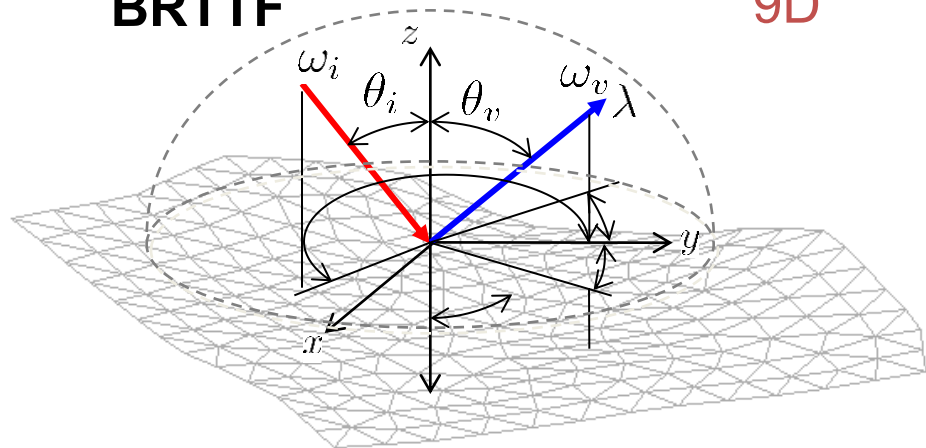
7D



$$Y_r = BTF(\lambda, x_i, y_i, \theta_i, \varphi_i, \theta_v, \varphi_v)$$

BRTTF

9D

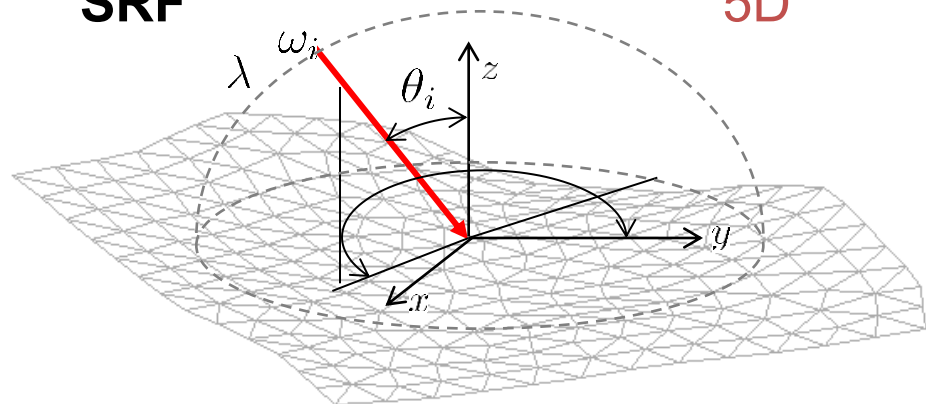


$$Y_r = BRTTF(\lambda, x, y, \theta_i, \varphi_i, \theta_v, \varphi_v, \theta_t, \varphi_t)$$

Textured Material Representation (2/2)

SRF

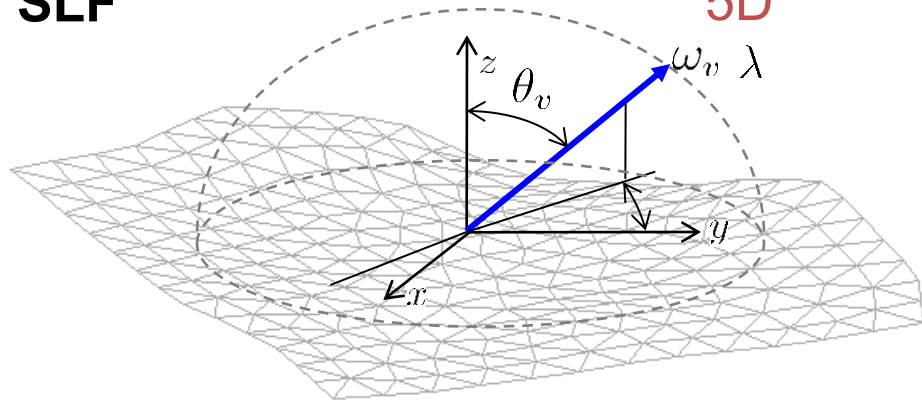
5D



$$Y_r(\theta_v, \varphi_v) = SRF(\lambda, x, y, \theta_i, \varphi_i)$$

SLF

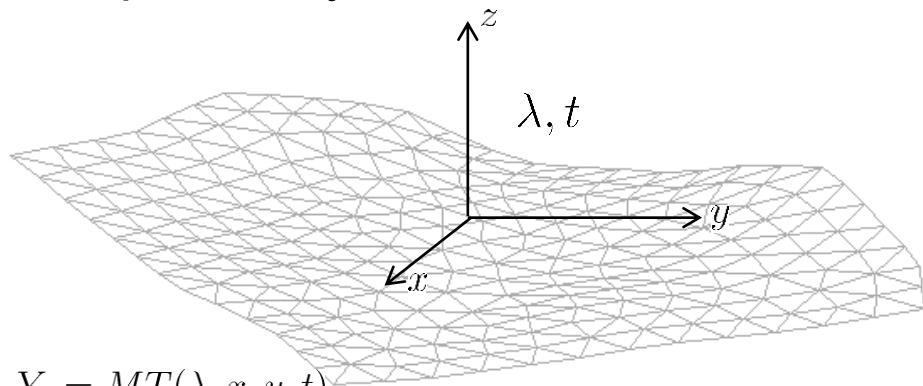
5D



$$Y_r(\theta_i, \varphi_i) = SLF(\lambda, x, y, \theta_v, \varphi_v)$$

Multispectral/dynamic texture

3/4D

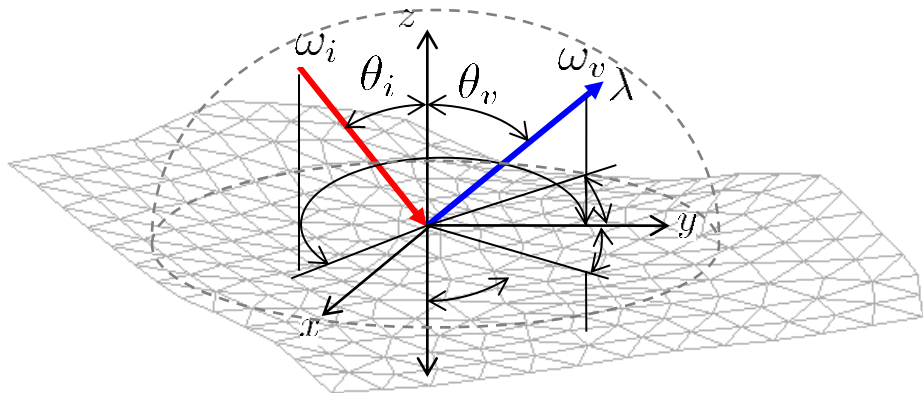


$$Y_r = MT(\lambda, x, y, t)$$

Homogeneous Materials Representations

BSDF

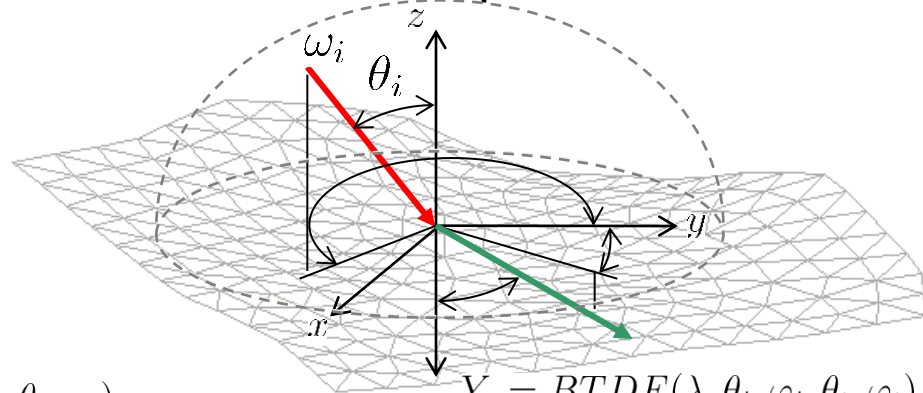
7D



$$Y_r = BSDF(\lambda, \omega_i, \varphi_i, \omega_v, \varphi_v, \theta_t, \varphi_t)$$

BTDF anisotropic

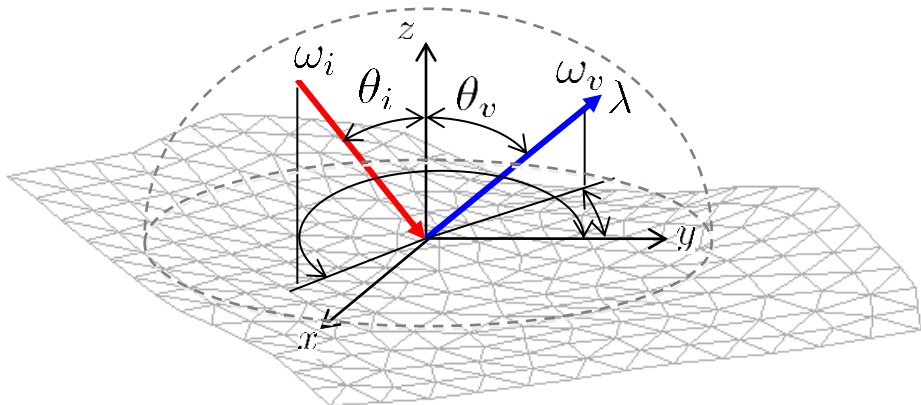
5D



$$Y_r = BTDF(\lambda, \theta_i, \varphi_i, \theta_t, \varphi_t)$$

BRDF anisotropic

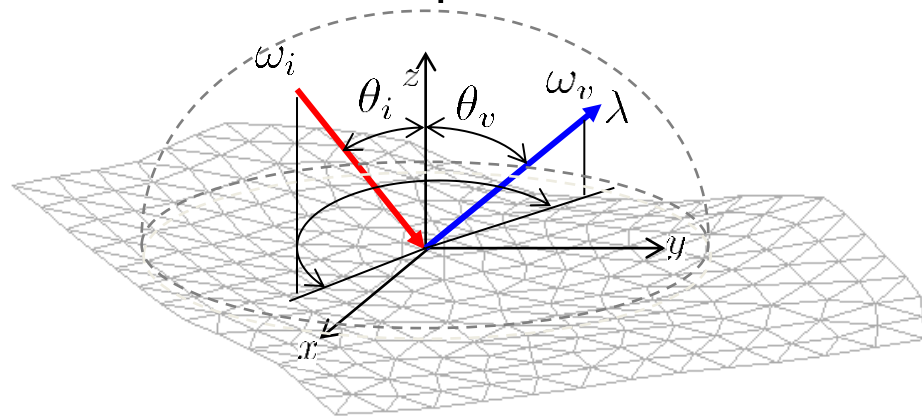
5D



$$Y_r = BRDF(\lambda, \theta_i, \varphi_i, \theta_v, \varphi_v)$$

BRDF isotropic

4D



$$Y_r = IBRDF(\lambda, \theta_i, \theta_v, |\varphi_i - \varphi_v|)$$

Materials Representations Examples



Appearance rendering

- Total **material radiance** $\Leftrightarrow$ emitted + reflected:

$$L_r(x, y, \omega_v) = L_{em}(x, y, \omega_v) + L_{rf}(x, y, \omega_v)$$

- General **rendering equation**

$$L_r(x, y, \omega_v) = \int_{\Omega_i} \underbrace{BRDF(\omega_i, \omega_v)}_{\text{Surface reflectance}} \underbrace{L_i(x, y, \omega_i)}_{\text{Illumination contribution}} \underbrace{\cos(\theta_i)}_{\text{Illuminated area foreshortening}} d\omega_i$$

Surface
reflectance

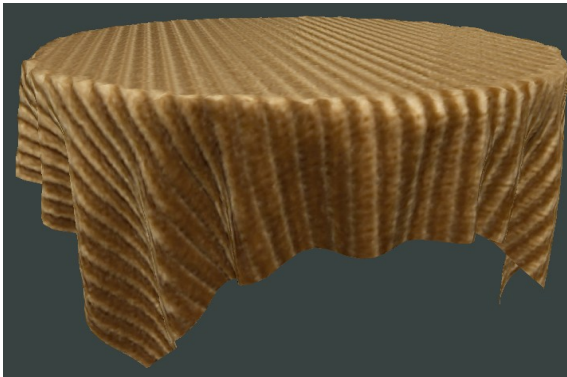
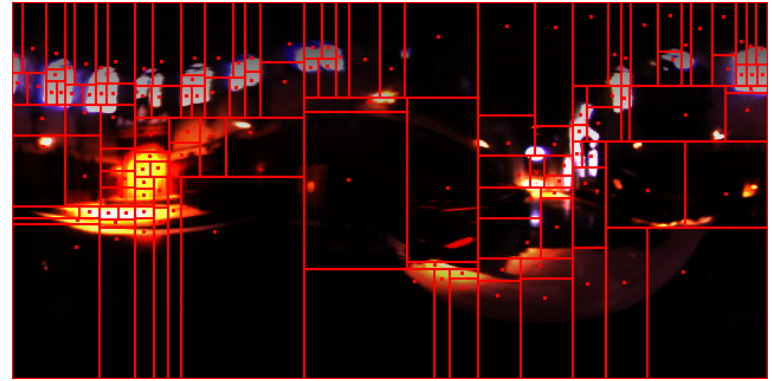
Illumination
contribution

Illuminated area
foreshortening

Illum./view
direction

Material visualization

- Global illumination – ray tracing approaches
- Approximation by set of pointlights



Radiometry

- Irradiance $E \left[\frac{W}{m^2} \right]$

$$E = \frac{d\Phi}{dA}$$

- Radiance $L \left[\frac{W}{m^2 \cdot sr} \right]$

$$L = \frac{d^2\Phi}{dA \cdot |\cos \theta| d\omega}$$

- BRDF

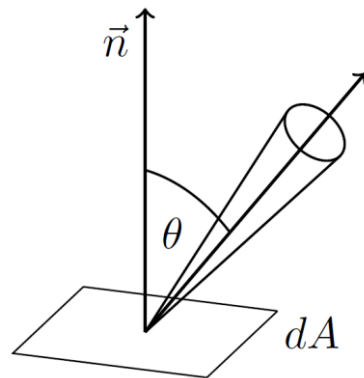
$$f(\vec{\omega}_{in}, \vec{\omega}_{out}, \lambda) = \frac{dL_{out}(\vec{\omega}_{out}, \lambda)}{dE(\vec{\omega}_{in}, \lambda)}$$

- Albedo

$$a(\vec{\omega}_{in}, \lambda) = \int_{\Omega} f(\vec{\omega}_{in}, \vec{\omega}_{out}, \lambda) \cdot \cos \theta_{out} d\vec{\omega}_{out}$$

- Mean albedo

$$a_{mean}(\lambda) = \frac{1}{\pi} \int_{\Omega} a(\vec{\omega}_{in}, \lambda) \cdot \cos \omega_{in} d\vec{\omega}_{in}$$



Radiometry

- Helmholtz reciprocity

$$\forall \vec{\omega}_{in}, \vec{\omega}_{out} \in \Omega : f(\vec{\omega}_{in}, \vec{\omega}_{out}, \lambda) = f(\vec{\omega}_{out}, \vec{\omega}_{in}, \lambda)$$

- Energy conservation

$$\forall \vec{\omega}_{in} \in \Omega : a(\vec{\omega}_{in}, \lambda) \leq 1$$

- Isotropy

$$\forall \vec{\omega}_{in} := (\phi_{in}, \theta_{in}), \vec{\omega}_{out} := (\phi_{out}, \theta_{out}) :$$

$$f((\phi_{in}, \theta_{in}), (\phi_{out}, \theta_{out}), \lambda) = f((0, \theta_{in}), (\phi_{out} - \phi_{in}, \theta_{out}), \lambda)$$

Bidirectional Reflectance Distribution Function

- Distribution of radiance reflected (L)

$$BRDF(\lambda, \theta_i, \phi_i, \theta_v, \phi_v) = \frac{dL_r(\lambda, \theta_v, \phi_v)}{\underbrace{L_i(\lambda, \theta_i, \phi_i) \cos \theta_i d\omega_i}_{\text{irradiance } E_i}}$$

- **Properties**

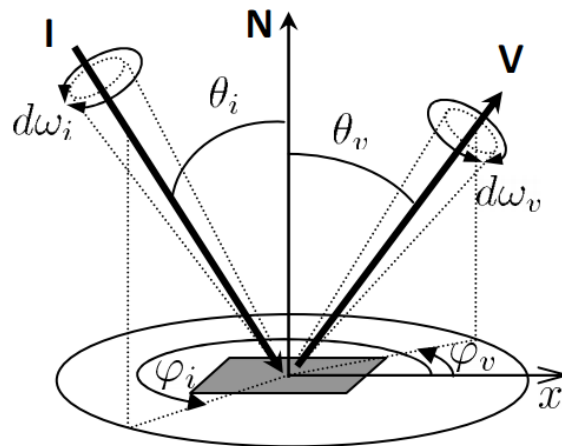
1. Illum./view directions reciprocity

- swapping source and sensor

2. Energy conservation

- portion of energy reflected to all directions has to be between 0 and 1

3. Non-negativity



Bidirectional Reflectance Distribution Function

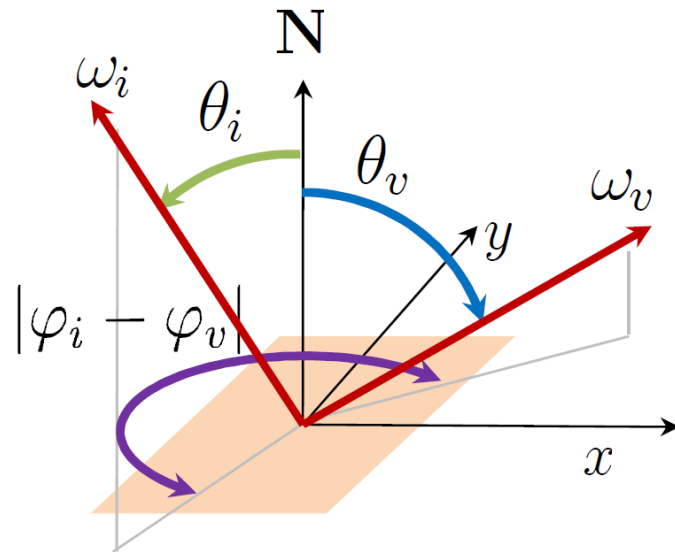
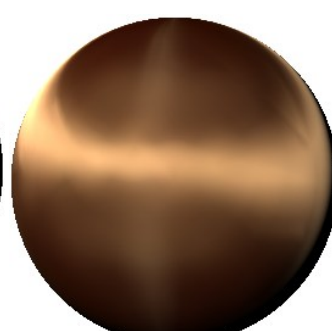
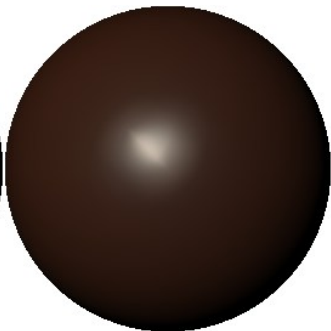
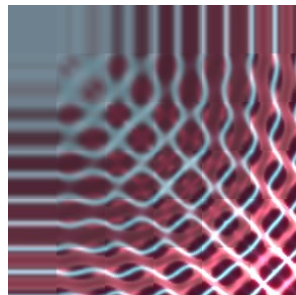
- Illumination and view dependent reflectance

$$BRDF(\lambda, \theta_i, \phi_i, \theta_v, \phi_v)$$

- Isotropic vs. anisotropic BRDF

$$B(\theta_i, \theta_v, |\varphi_i - \varphi_v|)$$

$$B(\theta_i, \varphi_i, \theta_v, \varphi_v)$$

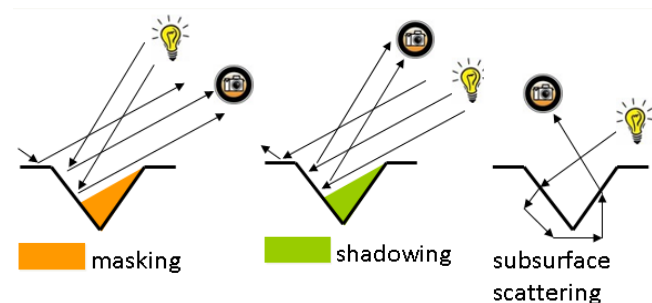


Bidirectional Texture Function

- BTF – Bidirectional Texture Function
- Illumination/view directions dependent texture

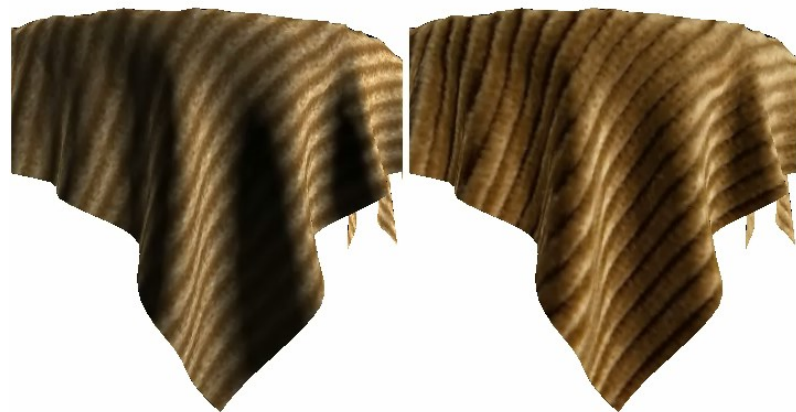
$$BTF(\lambda, x, y, \theta_i, \phi_i, \theta_v, \phi_v)$$

- Includes: inter-reflections, sub-surface scattering, local masking and shadowing
- One of the best practical representations of textured materials appearance
- Massive data $\Rightarrow$ thousands of images per material (GBs)
- Compression and modelling is inevitable



single texture

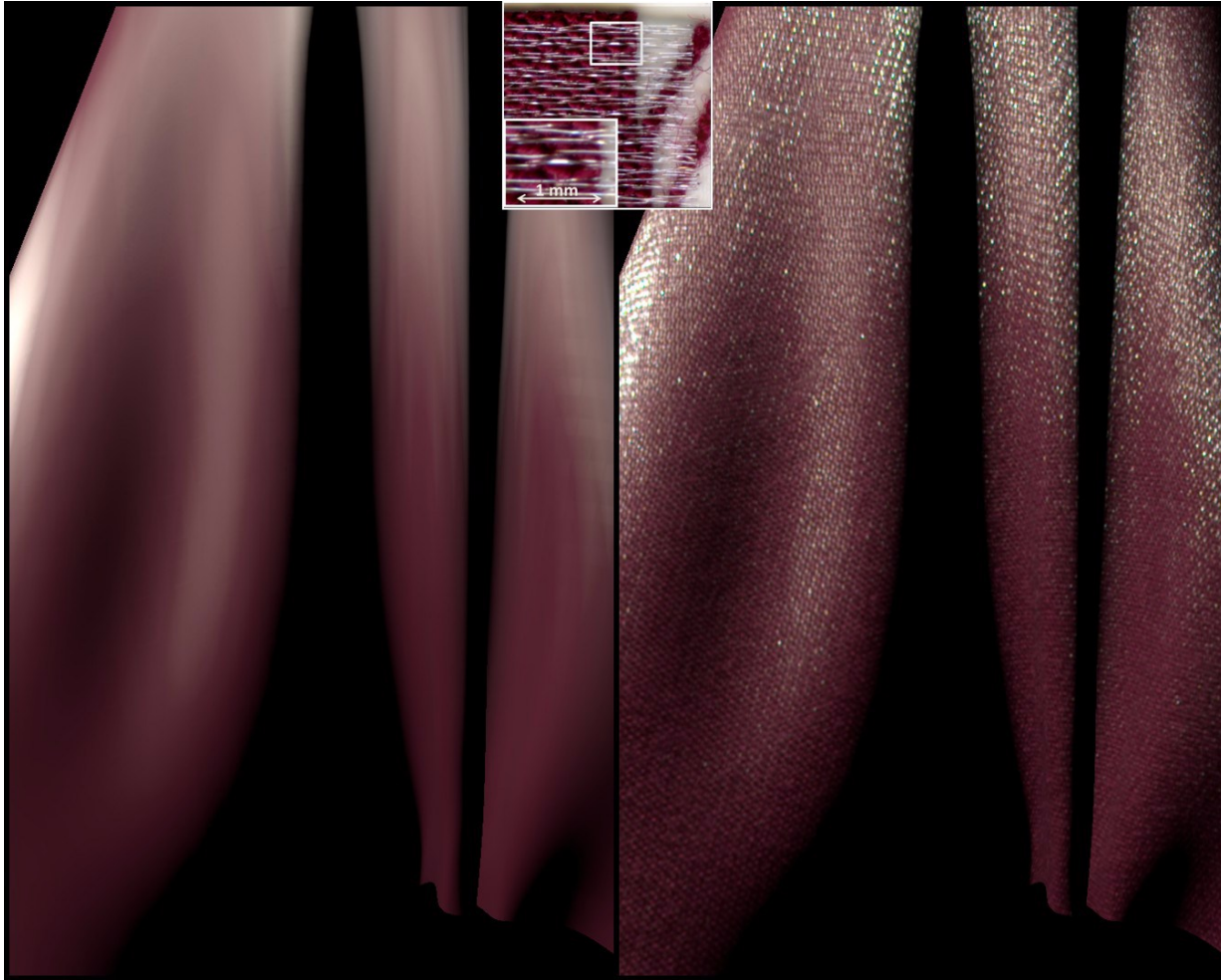
bidirectional texture function



Bidirectional Texture Function

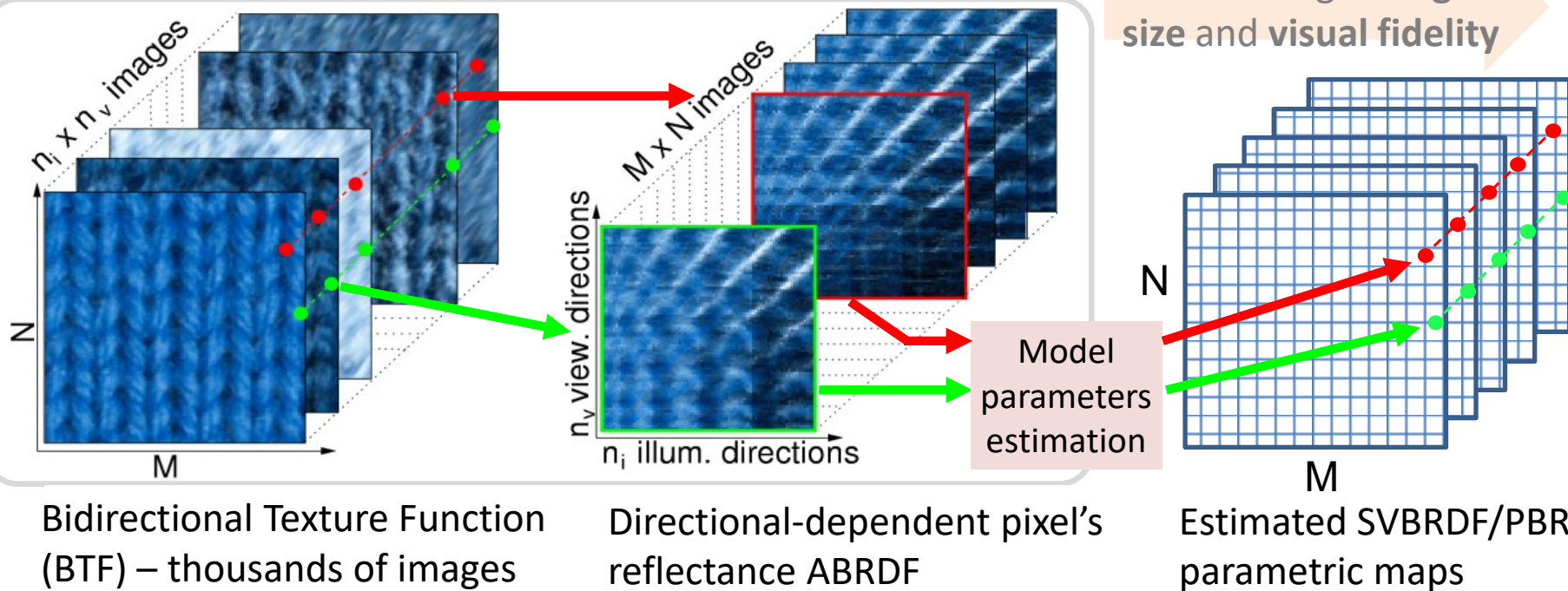


BRDF vs BTF



BTF vs Spatially-Varying BRDFs

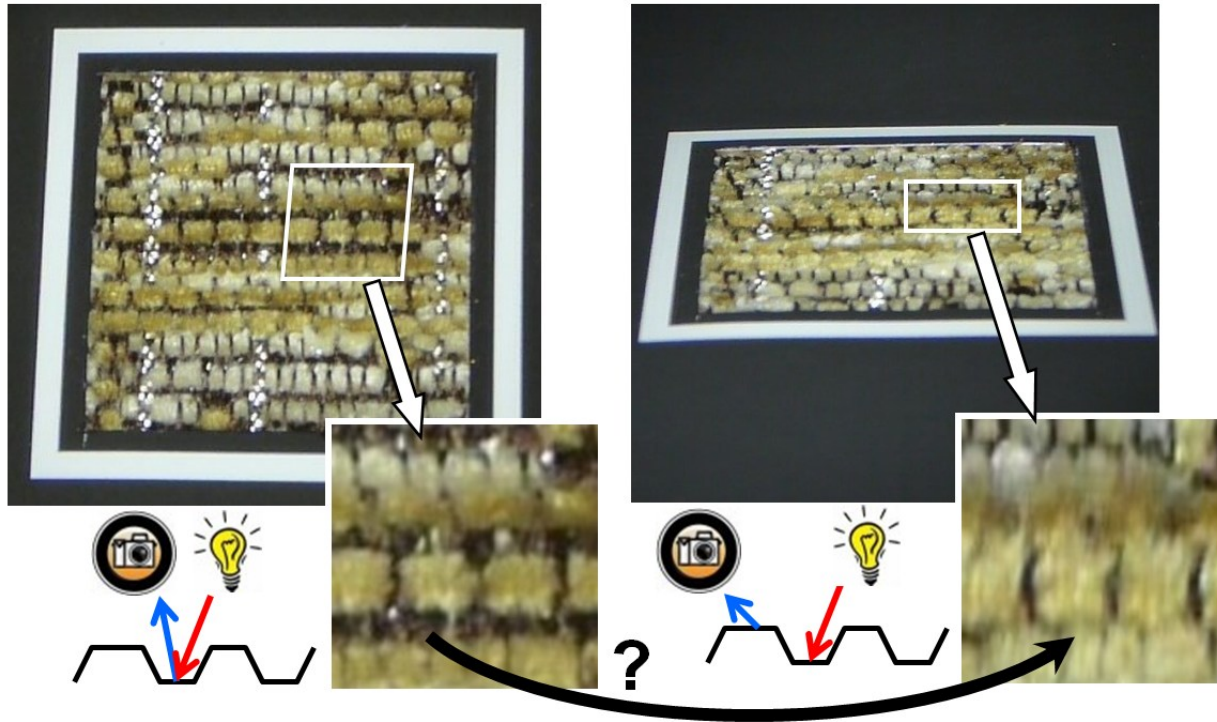
Captured spatio-angular reflectance



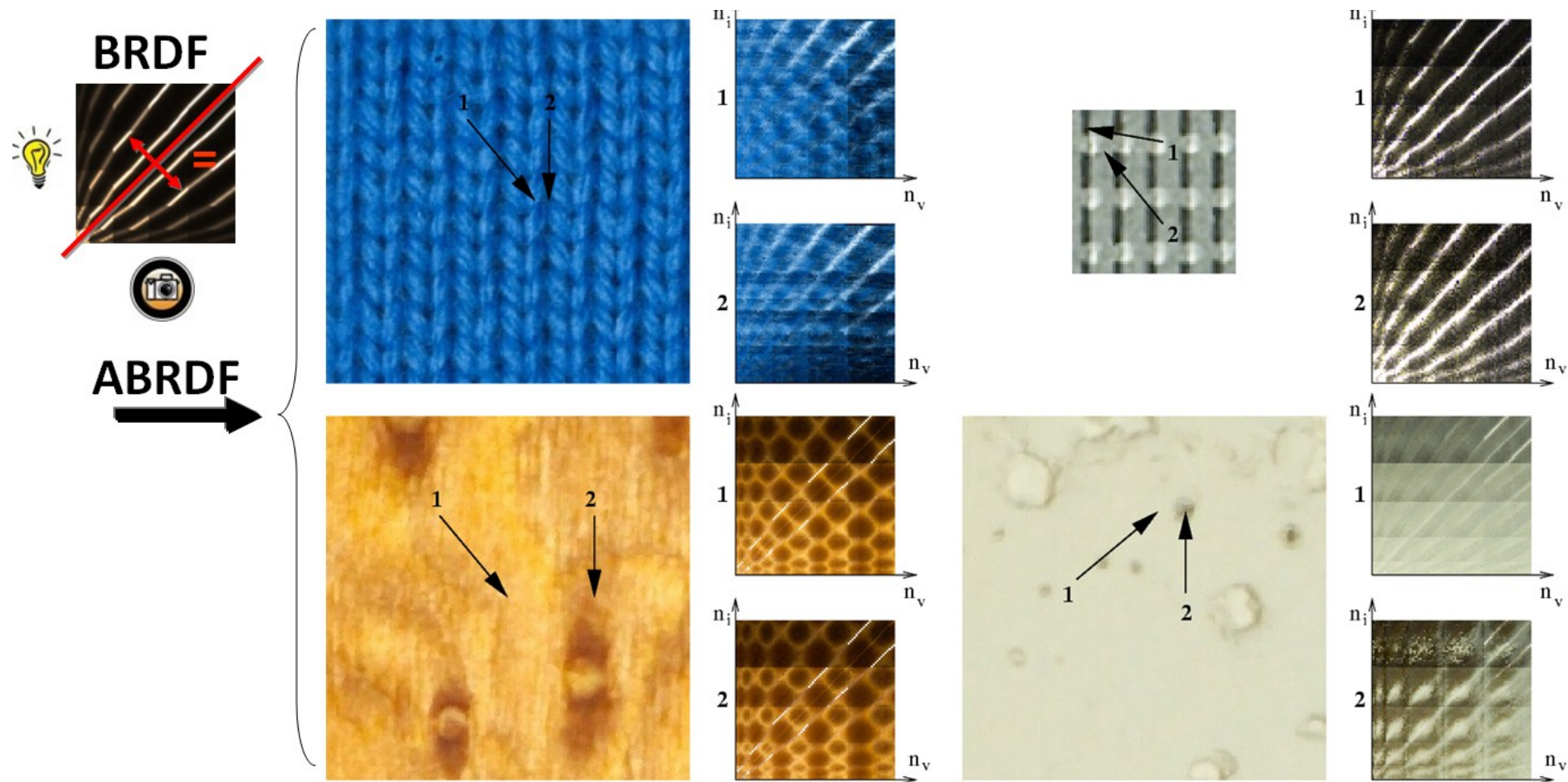
BTF vs Spatially-Varying BRDFs

Contrary to SVBRDF, BTF face problem with rough structure samples

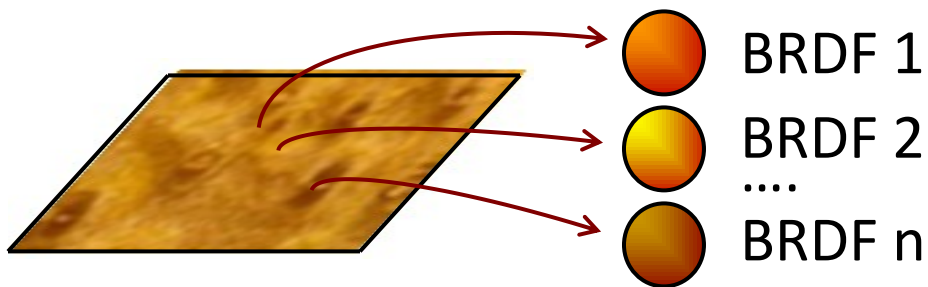
- Different views $\Rightarrow$ **Pixel-wise misalignment**, due to occlusion
- Solution $\Rightarrow$ process individual views separately



BRDF vs Pixel's Apparent BRDF



Spatially-Varying BRDFs



- **Restrictions**

- **Opaque surfaces** (no subsurface scattering)
- **Nearly flat surfaces** (no occlusions, masking $\Rightarrow$ reciprocity), no interaction between pixels

- **Advantages**

- No complicated light transport effects in material
- Allows spatial clustering $\Rightarrow$ each cluster single BRDF $\Rightarrow$ sparse effective measurements

$$SVBRDF(\lambda, x, y, \theta_i, \phi_i, \theta_v, \phi_v)$$

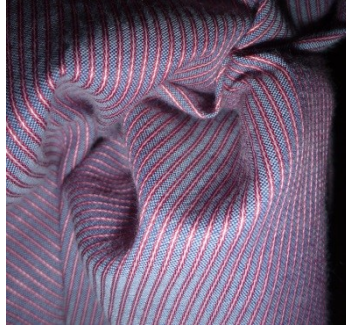
Challenges in material appearance

- Semitransparent materials



Challenges in material appearance

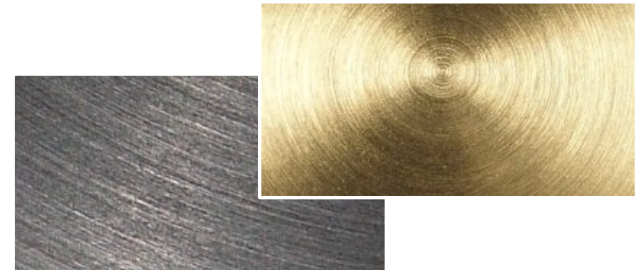
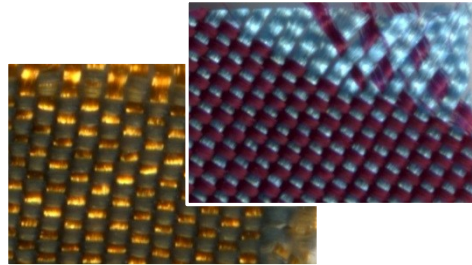
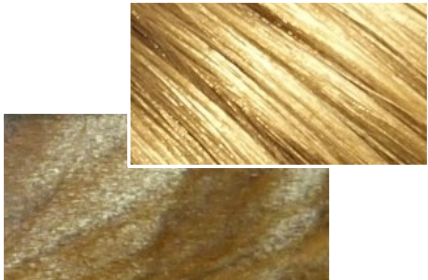
- Anisotropic materials



Directional fibers

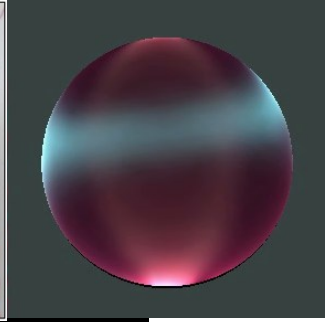
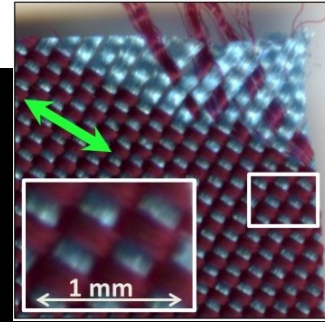
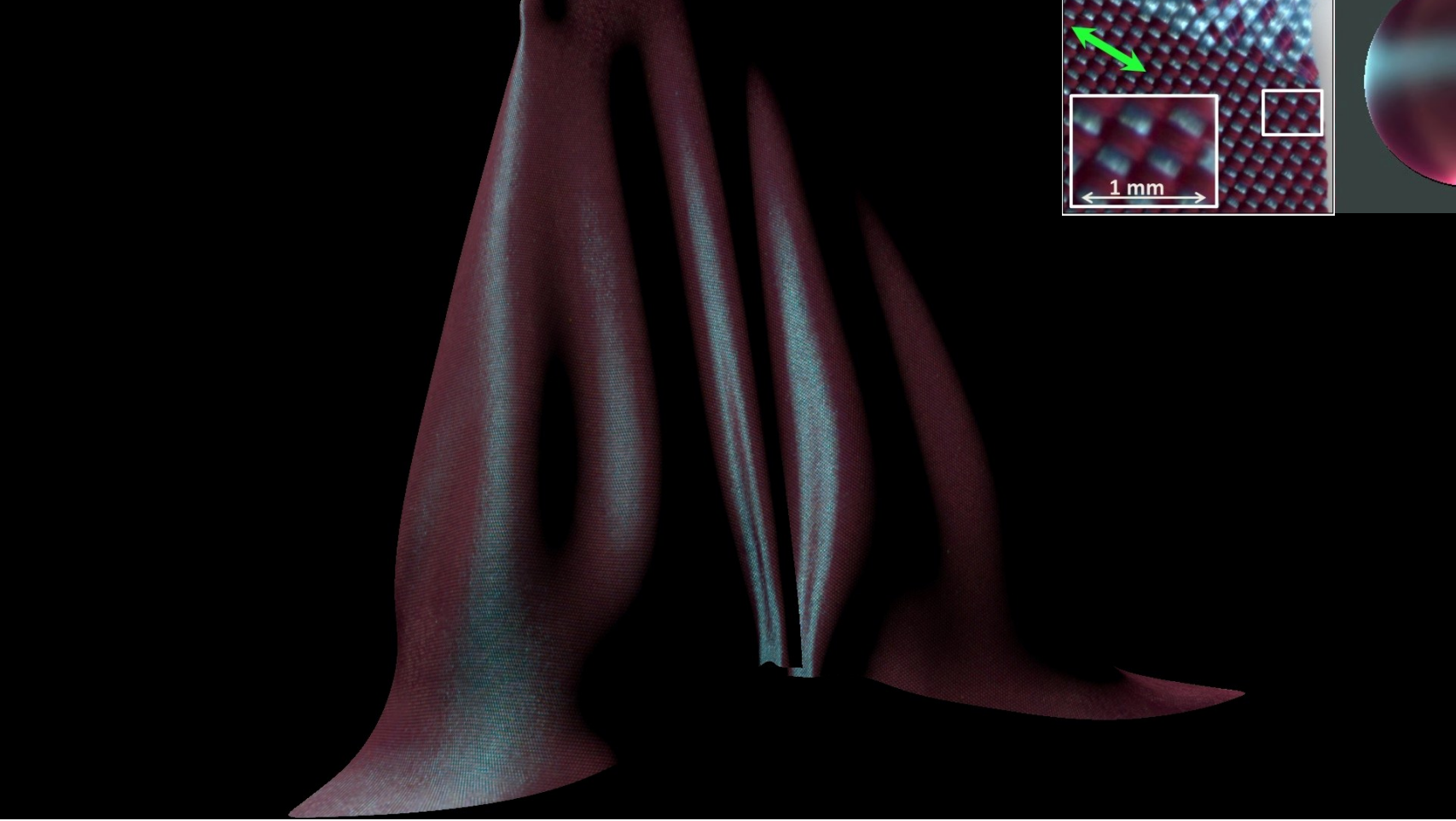
Weaving pattern

Surface finishing



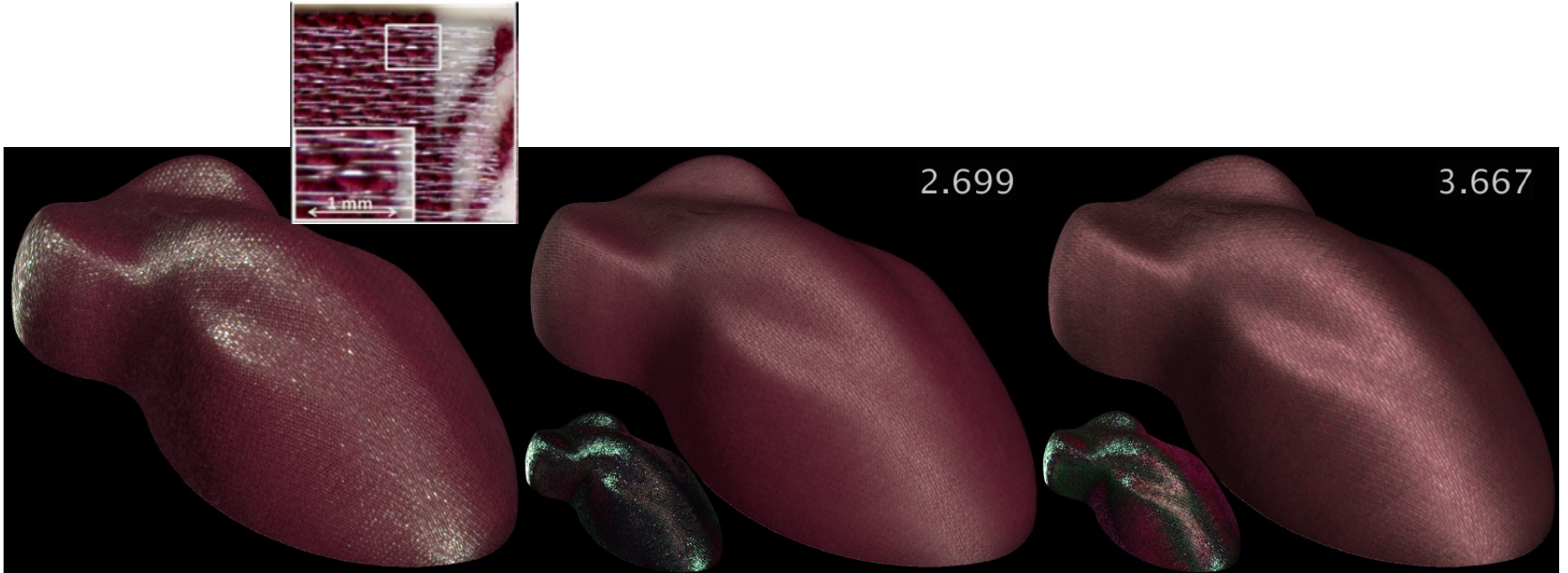
Challenges in material appearance

- Anisotropic materials



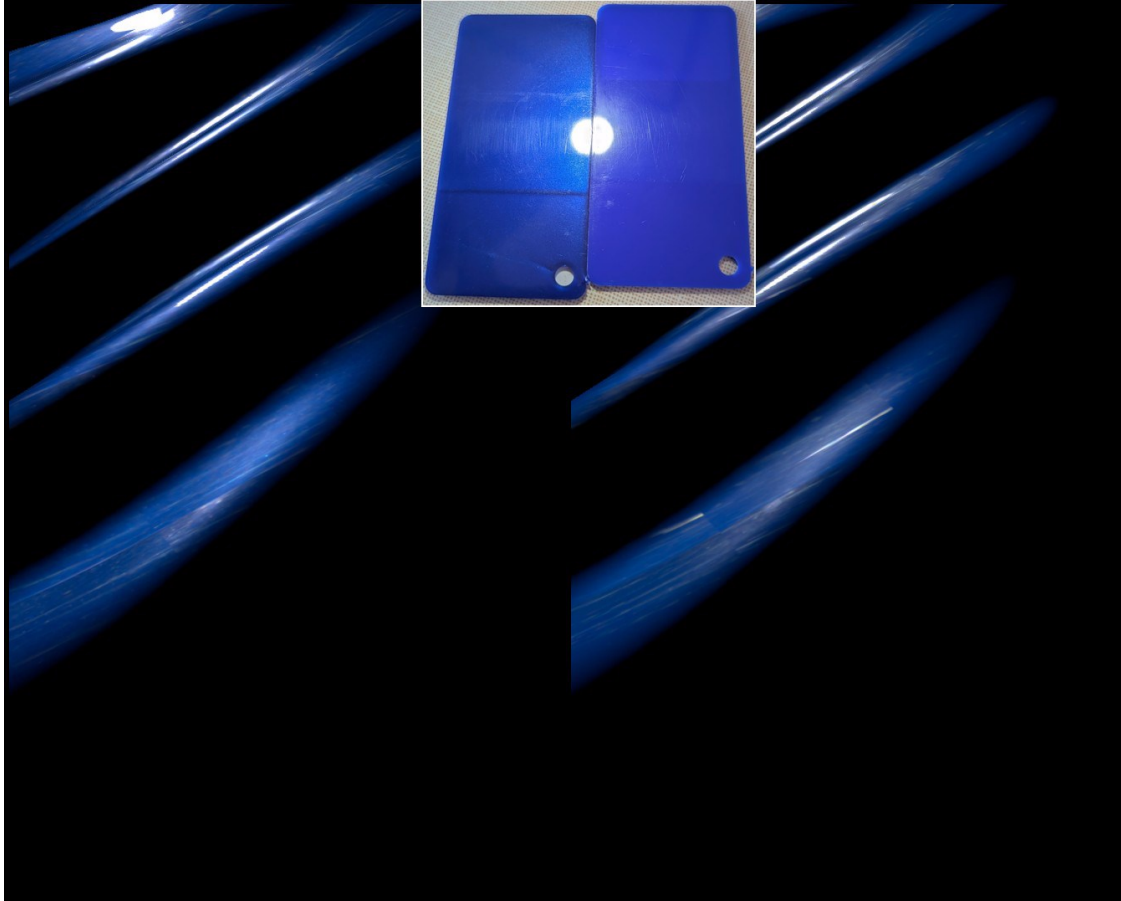
Challenges in material appearance

- Textured Specular materials



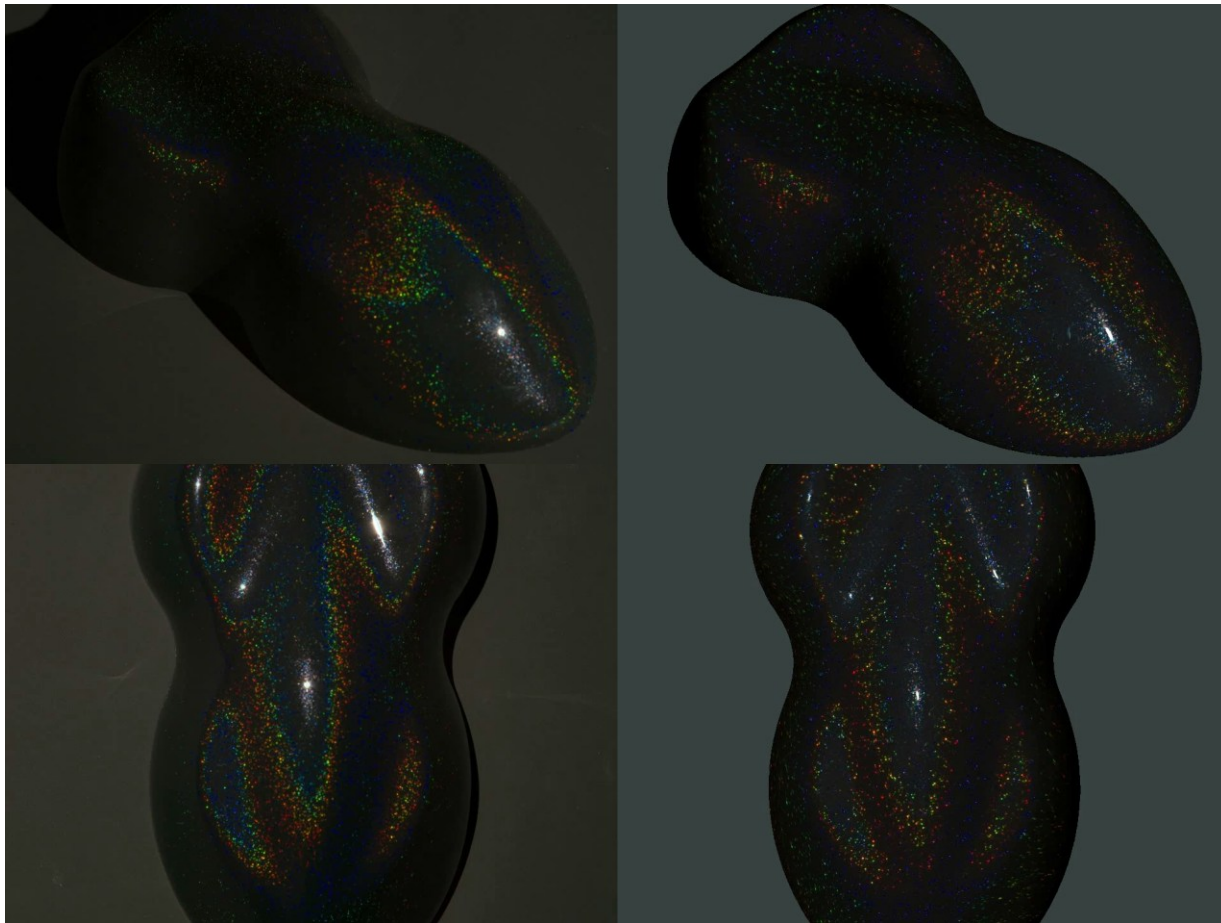
Challenges in material appearance

- Textured Specular materials



Challenges in material appearance

- Sparkle in effect coatings



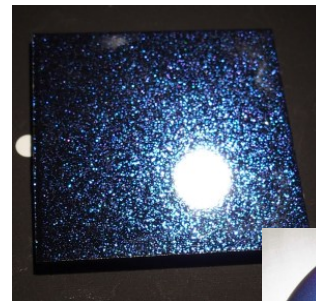
Challenges in material appearance

- Color-shifting materials



photographs

measured & rendered BTDF



Challenges in material appearance

- ... and their combinations

